

The Local-Chart Simplicial Complex as an Organizing Substrate for LPS

From the adaptive kNN graph to the nerve of the chart cover — an organizational (not topological) scaffold for refining conditional-expectation and local-covariance estimation

Internal design note — LPS / geosmooth program

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Abstract

Every estimator in the LPS program currently sits on top of a graph that is built *before, and independently of*, the estimation problem: an adaptive-radius k -nearest-neighbor graph. This note argues that once we commit to *local chart* models, the natural combinatorial substrate is not that input graph but the *nerve of the chart cover* — the simplicial complex whose vertices are the charts and whose simplices record which charts overlap. The thesis is deliberately organizational, not topological: we are not interested in reconstructing the data’s homotopy type, only in a faithful bookkeeping of *which local estimates should agree, and how strongly*, so that conditional expectations and local covariances can be refined by gluing. Seen this way, the program already uses the shadow of this object — PS-LPS’s overlap coupling and the chart-aware association layer’s overlap weights are exactly the nerve’s 1-skeleton. Making the nerve native buys three things a kNN graph cannot give: variable chart sizes (an adaptive cover), higher-order overlaps (constraints where three or more charts meet), and a substrate whose quality is judged by held-out estimation accuracy rather than fixed by an a-priori rule. We lay out what each LPS layer gains, the one criterion that makes a complex “good” for our purposes (a statistical one), the compositional case where part of the complex is known for free, the real costs, and a minimal-first path keyed to the synthetic generators of the implementation notes.

On the figures. All figures are original schematics drawn for this note.

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1 The substrate, and a deliberate restriction of ambition

The LPS pipeline builds, in order: a neighbor graph on the samples; a local chart at each anchor (local PCA, with an estimated local dimension); a local polynomial fit on that chart; and, in PS-LPS and the chart-aware association layer, a coupling that ties *overlapping* charts together. The first object — the adaptive-radius kNN graph — is built by a generic proximity rule that knows nothing about the regression surface or the covariance structure we are trying to estimate. It is scaffolding.

The thesis of this note is that the charts themselves induce a better substrate. A collection of charts is a *cover* of the data: each chart’s support is a subset U_c of samples, and together the U_c blanket the cloud. The combinatorial object that records how a cover fits together is its *nerve*.

Definition 1.1 (Nerve of the chart cover). Let $\{U_c\}_c$ be the supports of the local charts. The nerve $\mathcal{N}$ is the simplicial complex with one vertex per chart c , an edge $\{c, c'\}$ whenever $U_c \cap U_{c'} \neq \emptyset$, a triangle $\{c, c', c''\}$ whenever $U_c \cap U_{c'} \cap U_{c''} \neq \emptyset$, and so on. Each simplex carries the size of the corresponding overlap as a weight.

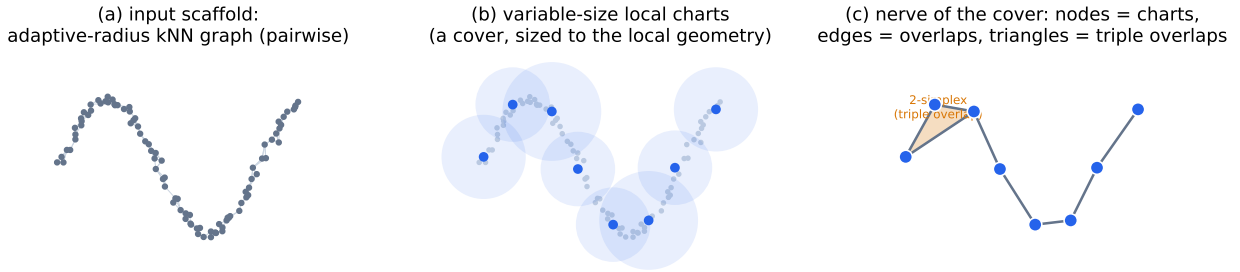


Figure 1: The shift in substrate. (a) The input scaffold: an adaptive-radius kNN graph, pairwise and fixed before estimation. (b) The charts form a *cover* whose pieces can vary in size with the local geometry. (c) The nerve of that cover: charts are nodes, an edge marks a pairwise overlap, and a filled triangle marks a place where three charts overlap at once — structure a pairwise graph cannot record.

We must be precise about what we want from $\mathcal{N}$, because the word “nerve” comes loaded with topology. The classical reason to care about a nerve is the Nerve Theorem: for a sufficiently nice (“good”) cover, the nerve is homotopy equivalent to the union of the cover, so one can read the space’s topology off a finite complex. *We are explicitly not relying on that.* The goal here is biology, and the use is organizational: $\mathcal{N}$ is a faithful ledger of which local estimates overlap, and therefore of which ones should be made to agree and with what strength. Whether $\mathcal{N}$ reproduces the data’s homology is beside the point; what matters is whether gluing local estimates over $\mathcal{N}$ produces better conditional expectations and covariances. That reframing — from a topological criterion to a statistical one — runs through the whole note.

In plain terms Think of the charts as overlapping photographs that together cover a landscape. The “nerve” is just the index card that says which photos overlap, and which three overlap at a corner. We are not trying to rebuild the landscape’s shape from the index card; we are using the card to know which photos must be stitched to agree where they overlap — so the stitched panorama (our estimate of the surface) is seamless.

2 The program already uses the nerve’s shadow

This is not an exotic import. Two pieces of the existing program are, up to naming, already defined on $\mathcal{N}$.

- **PS-LPS** couples two charts when their supports share points and penalizes disagreement of their predictions on the shared points. “Charts that share support points” is precisely an *edge* of $\mathcal{N}$; the shared points are the overlap that weights it. PS-LPS is a synchronization defined on the nerve’s 1-skeleton.
- **The chart-aware association layer** fuses neighboring charts’ covariance/precision blocks with an overlap weight $\zeta_{cc'} = \sum_j \min(w_{cj}, w_{c'j})$ — again an edge weight of $\mathcal{N}$, measuring how much two charts overlap.

So the program is already doing nerve-based gluing; it just derives the adjacency from the kNN graph and stops at pairwise edges. The proposal is to (i) derive $\mathcal{N}$ *natively* from the charts (so it adapts as the charts adapt), and (ii) use more of it than the bare 1-skeleton.

3 Why the nerve organizes better than any kNN graph

Three differences matter, and all three are in service of estimation, not topology.

Variable chart size = an adaptive cover. A kNN graph fixes the arity (or radius) of every neighborhood before looking at the problem. Charts need not: a chart can be large where the data is sparse or locally one-dimensional and small where it is dense or higher-dimensional. The nerve of a variable-size cover encodes this adaptivity directly — two regions are coupled exactly when their *charts* (sized to the local geometry) overlap, not when an arbitrary fixed k says so.

Higher-order overlaps = constraints a graph cannot express. Where three charts overlap, the right object is a single three-way agreement, not three separate pairwise ones. A pairwise graph can only ever say “ a agrees with b , b with c , a with c ”; it cannot say “ a , b , c agree on their common region.” The latter is a genuinely different (and at junctions, more appropriate) constraint, and it is exactly a 2-simplex of $\mathcal{N}$ (Figure 2, right). This is the same reason the clique/Rips complex induced by a kNN graph — which manufactures higher simplices from pairwise edges — is a poorer record of the cover than the nerve, which records the actual multi-way intersections.

The substrate becomes part of the model. Because $\mathcal{N}$ is built from the charts, and the charts are chosen to fit the estimation problem, the substrate is no longer fixed input: it is selected, and can be judged by how well estimates glued over it predict held-out data (Section 5).

A caution worth stating None of this requires the cover to be “good” in the topological sense. A kNN graph and the nerve will sometimes disagree about whether two regions should be coupled, and near a stratum boundary the charts may overlap in ways that would *not* make a topologically faithful nerve. For our purposes that is fine: an extra or missing edge changes which estimates are tied together and how strongly — a modeling choice to be validated by accuracy, not a topological error. We keep the nerve as a statistical ledger and let held-out error referee it.

4 What each LPS layer gains

4.1 Conditional expectation (PS-LPS)

Moving PS-LPS onto the native nerve changes three things. *Adaptivity*: the coupling graph is the nerve of variable-size charts, so two anchors are synchronized exactly when their (geometry-sized) charts overlap, with strength set by the overlap. *Higher order*: on a 2-simplex, synchronize the three charts’ predictions on their common overlap as a single constraint — the natural thing at a junction where three local models meet (Figure 2). *Honest weights*: edge and triangle weights come from actual intersection sizes, not from a kNN kernel evaluated at fixed arity.

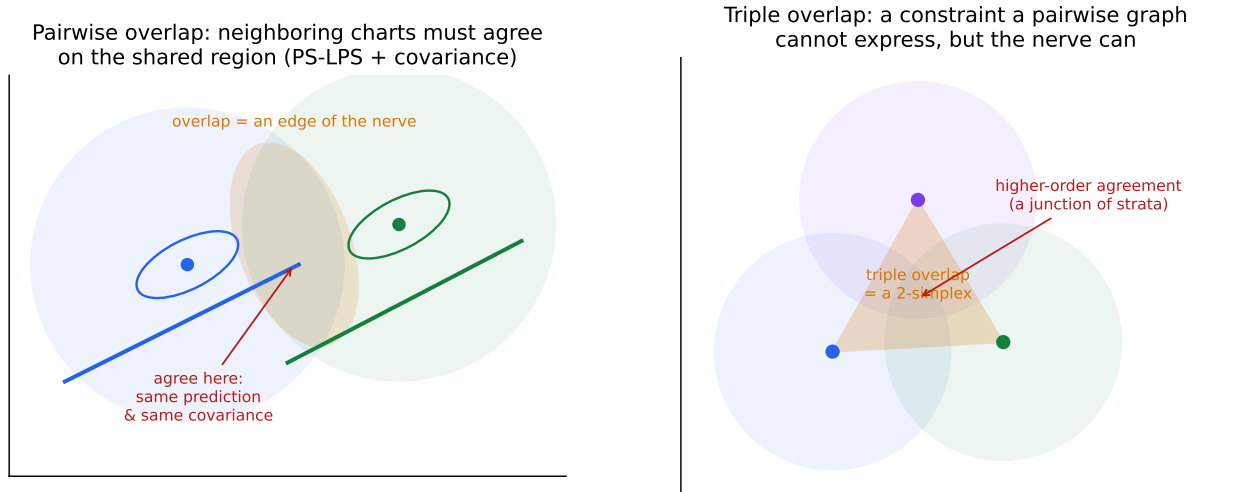


Figure 2: Gluing on overlaps. Left: where two charts overlap (an edge of $\mathcal{N}$) their local fits — both the predicted surface and the local covariance — should agree on the shared region. Right: where three charts overlap (a 2-simplex) there is a three-way agreement that a pairwise graph cannot express. These overlaps are exactly where, and how strongly, local estimates are tied together.

4.2 Local covariance and association

The chart-aware layer estimates a covariance/precision per chart and fuses neighbors. The overlap of two charts is precisely the set of samples on which their two covariance estimates are looking at the same data — so $\mathcal{N}$ tells us *where* the covariance estimates should agree and *how strongly* (the overlap weight). Variable chart size lets the covariance model widen its support where data is scarce. And the higher simplices give three-way covariance agreement at junctions, where the support layer (which taxa co-occur) reorganizes.

4.3 The local-dimension / chart-size field

The dimension field of the implementation notes minimizes $\sum_i c_i(d_i) + \lambda \sum_{(i,j)} w_{ij} \rho(d_i, d_j)$ over a graph; that graph should be the nerve 1-skeleton, with weights w_{ij} from overlap sizes (dimension- and size-aware) rather than kNN adjacency. More: the chart *size* is itself a field — larger charts average more (lower variance, more bias) — and it should be co-regularized with the dimension over $\mathcal{N}$, with the same edge-preserving (ℓ_1) penalty so that size, like dimension, stays constant within a stratum and jumps at boundaries.

4.4 One objective over the complex

Attach to each chart its local data — predicted surface, covariance, dimension, size. The nerve records which charts overlap. The global estimate minimizes

$$\sum_c \underbrace{L_c(\text{local fit})}_{\text{fidelity}} + \sum_{\sigma \in \mathcal{N}} \omega_\sigma \underbrace{D_\sigma(\text{disagreement on } \bigcap_{c \in \sigma} U_c)}_{\text{gluing}},$$

where σ ranges over the simplices (edges, triangles, ...) of $\mathcal{N}$ and ω_σ is the overlap weight. PS-LPS prediction synchronization, the covariance fusion, and the dimension/size field are all special cases — each is a particular choice of what the local data is and how disagreement on an overlap is measured. (Formally this is a cochain energy of a cellular sheaf over $\mathcal{N}$; we use that only as the name for “make the local data agree on overlaps,” not for any topological content.)

In plain terms Every layer of the program is, underneath, the same sentence: each chart fits its own little patch of data, and overlapping charts are pulled toward agreeing where they overlap. The nerve is just the precise list of overlaps — including the three-way ones — and how big each is. Writing the program against that list, instead of against a generic neighbor graph, makes every layer use the same, geometry-aware bookkeeping.

5 What makes a complex “good” here: a statistical criterion

If we are not chasing topology, what tells us we have the *right* complex? The answer is the only one that matters for the program: a complex is good to the extent that estimates glued over it predict held-out data better. The overlap structure is a modeling choice — which local estimates to tie together, and how strongly — and modeling choices in this program are settled by held-out conditional-expectation accuracy (the same rule the implementation notes use to select the dimension field’s λ). This dissolves the would-be chicken-and-egg of “you need the geometry to build the cover, and the cover to estimate the geometry”: we do not need the cover to be correct in any absolute sense, only useful, and usefulness is measurable.

It also turns the user’s original observation — that the substrate should be *derived from* the chart models — into a concrete loop (Figure 3). Start from charts (each with a size and a local dimension); take the nerve of their cover; glue the local estimates over it; score the result by held-out prediction error; and let that score re-size and re-dimension the charts. A fixed point of this loop is a self-consistent atlas whose overlaps were chosen by the estimation problem itself. In practice one would run a few rounds rather than seek an exact fixed point; whether the loop converges, and to what, is an open question (Section 8), but each round is well-defined and each is judged by accuracy.

The estimation problem chooses the substrate: charts $\rightarrow$ nerve $\rightarrow$ glued estimates $\rightarrow$ refine, guided by held-out conditional-expectation accuracy (not by topology).

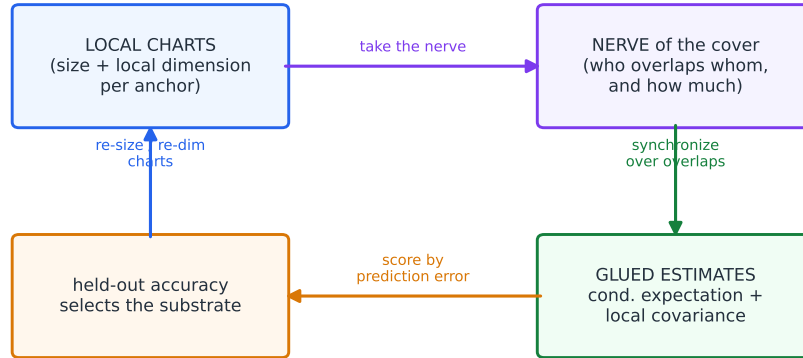


Figure 3: The estimation problem chooses the substrate. Charts (with their sizes and local dimensions) induce a cover; its nerve says who overlaps; local estimates are glued over the nerve into conditional-expectation and covariance fields; held-out accuracy scores the result and feeds back to re-size and re-dimension the charts. The cover is selected, not fixed.

6 The compositional case: part of the nerve is known for free

For 16S relative-abundance data the nerve is not entirely something to estimate — much of its coarse combinatorics is handed to us by the simplex. A sample’s support (which phylotypes are present) places it on a face; two charts on different faces F_A and $F_{A'}$ can overlap only through their shared sub-face, and which faces are adjacent is fixed by the face lattice (the support patterns enumerated by experiment E1 of the implementation notes). So the cross-face part of $\mathcal{N}$ — which strata are even capable of being coupled — is a known, finite poset, not a fitted object.

What remains to estimate is the *within-face* part: among samples sharing a support, which charts overlap, at what sizes, and where the effective dimension drops (the gaps measured by E2). This is exactly where the variable-size, dimension-aware nerve earns its keep, and it scopes the work honestly: the expensive, learned part of the substrate lives inside the strata and at their soft boundaries (near-ties, near-zeros), while the skeleton across strata is read off the zero pattern.

In plain terms In microbiome data, the big pieces of the “who can overlap whom” map are free: two samples made of completely different microbes simply do not share a chart. What we actually have to learn is the fine structure *within* a group of samples that share the same microbes — how their charts overlap and how many directions they really spread in.

7 Costs, and a minimal-first path

Costs, stated plainly. Higher simplices are combinatorially heavier: the number of triple overlaps can be large, so the complex is truncated at dimension two or three and built only among nearby charts — still a sparse object, but a real implementation cost. Variable chart size adds a selection problem (more adaptivity means more to choose and more variance), which is why the size field must be regularized jointly with the dimension over $\mathcal{N}$. And the assembly and cross-validation bookkeeping must be re-expressed against the nerve rather than the kNN graph; because PS-LPS

already assembles its system from chart overlaps, this is a generalization of existing code, not a rewrite.

Minimal-first. The idea should be earned one measurable step at a time, on the synthetic generators of the implementation notes (where the true local-dimension field, and hence the intended overlaps, are known):

1. Replace PS-LPS’s pairwise overlap graph with the nerve 1-skeleton of *variable-size* charts. Test whether this alone improves held-out Truth-RMSE on a stratified generator. This is a small change with an immediate adaptivity payoff.
2. Add 2-simplices and higher-order synchronization on triple overlaps. Test whether they measurably help — the hypothesis is that they matter only near junctions of strata.
3. Co-regularize chart *size* and chart *dimension* as fields over $\mathcal{N}$ with an ℓ_1 penalty, so both stay constant within a stratum and jump at boundaries.
4. Only if steps 1–3 pay off, lift prediction, covariance, and dimension into the single cochain objective of Section 4.

At every step the substrate is judged by held-out estimation accuracy, never by a topological or intrinsic-geometry score.

8 Open questions

- **Sizing for a statistically good cover.** How should chart sizes be chosen so that the nerve is a useful ledger — jointly with the local dimension, and judged by accuracy? This is the substantive modeling question.
- **The refinement loop.** Does the charts $\rightarrow$ nerve $\rightarrow$ glue $\rightarrow$ refine loop converge, and is the fixed point meaningful, or is a fixed number of rounds the right thing?
- **Does higher order pay?** The empirical crux: do 2-simplex (triple) constraints measurably improve estimates over the 1-skeleton, and only at junctions, as conjectured?
- **Cheap truncation.** What is the cheapest construction of the dimension- ≤ 2 nerve that keeps the overlaps that matter for estimation?
- **Relation to existing machinery.** How does a nerve-based assembly sit with the metric-graph and graph-trend-filtering tooling already in `geosmooth` — can the same solvers be reused on the nerve 1-skeleton?

These are the right next questions precisely because they are all answerable by experiment on synthetic data with known truth, and then by held-out accuracy on the VALENCIA-shaped generators — which keeps the whole idea, ambitious as it is, anchored to biology and to measurable estimation gains rather than to topology.

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